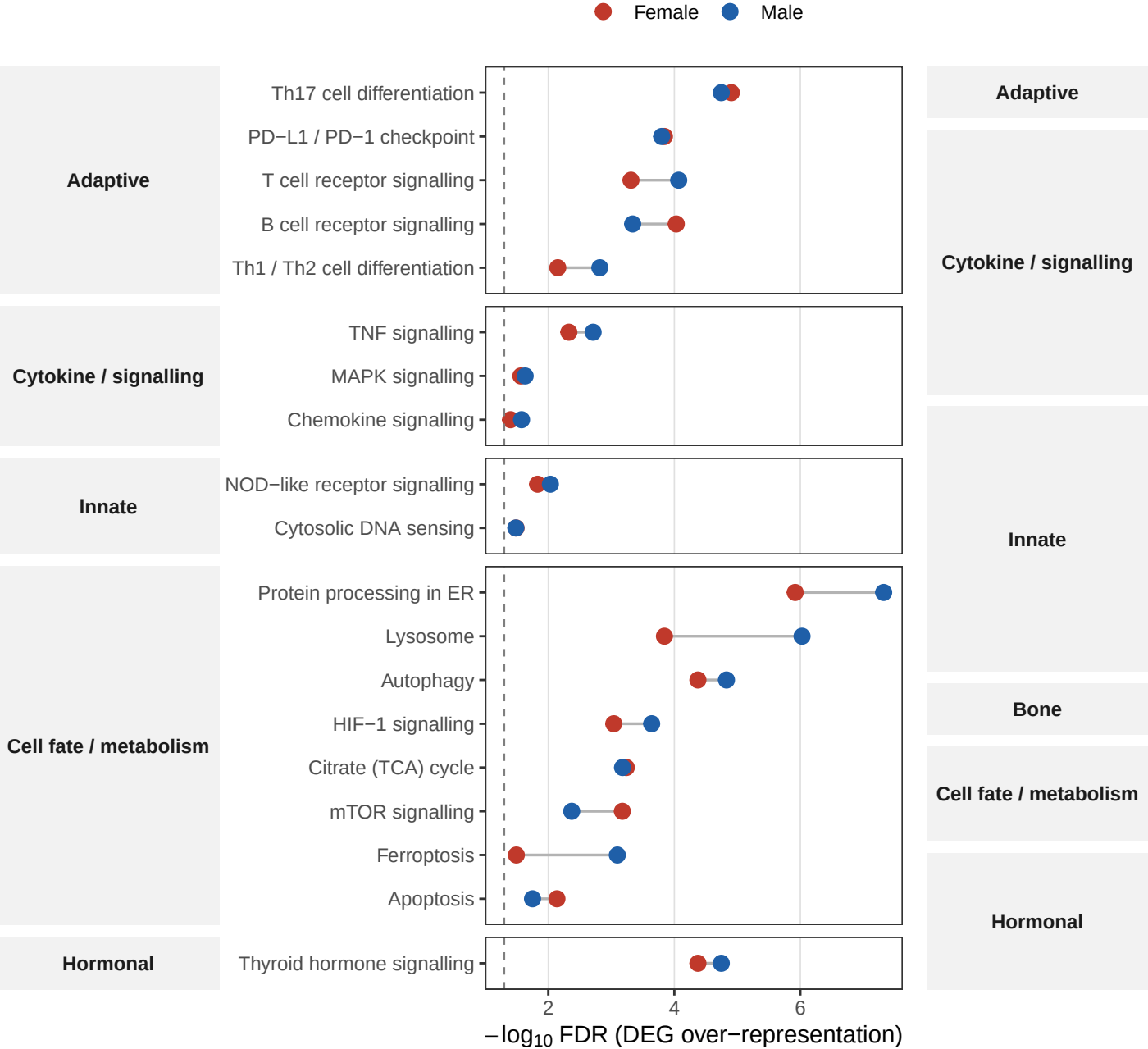
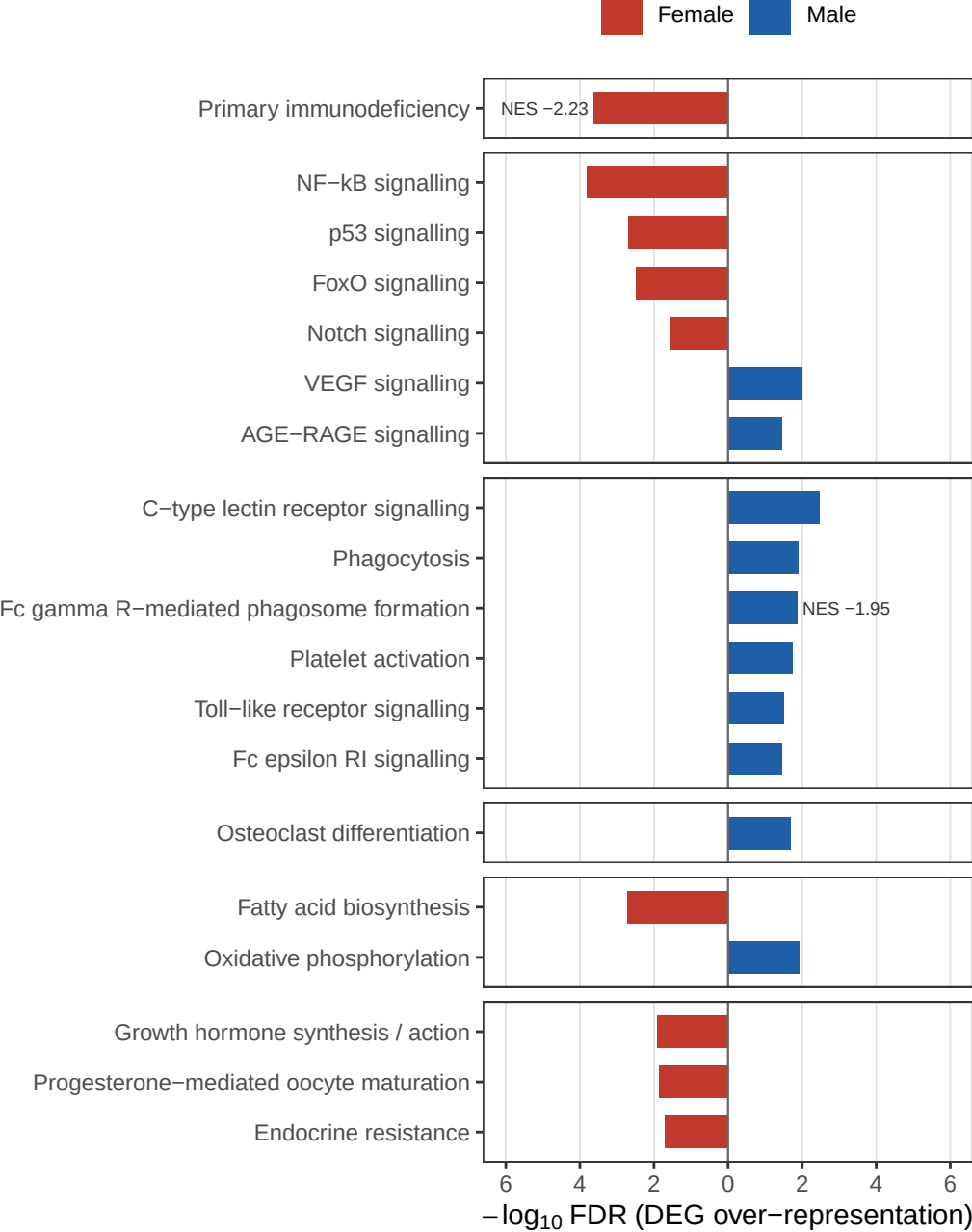


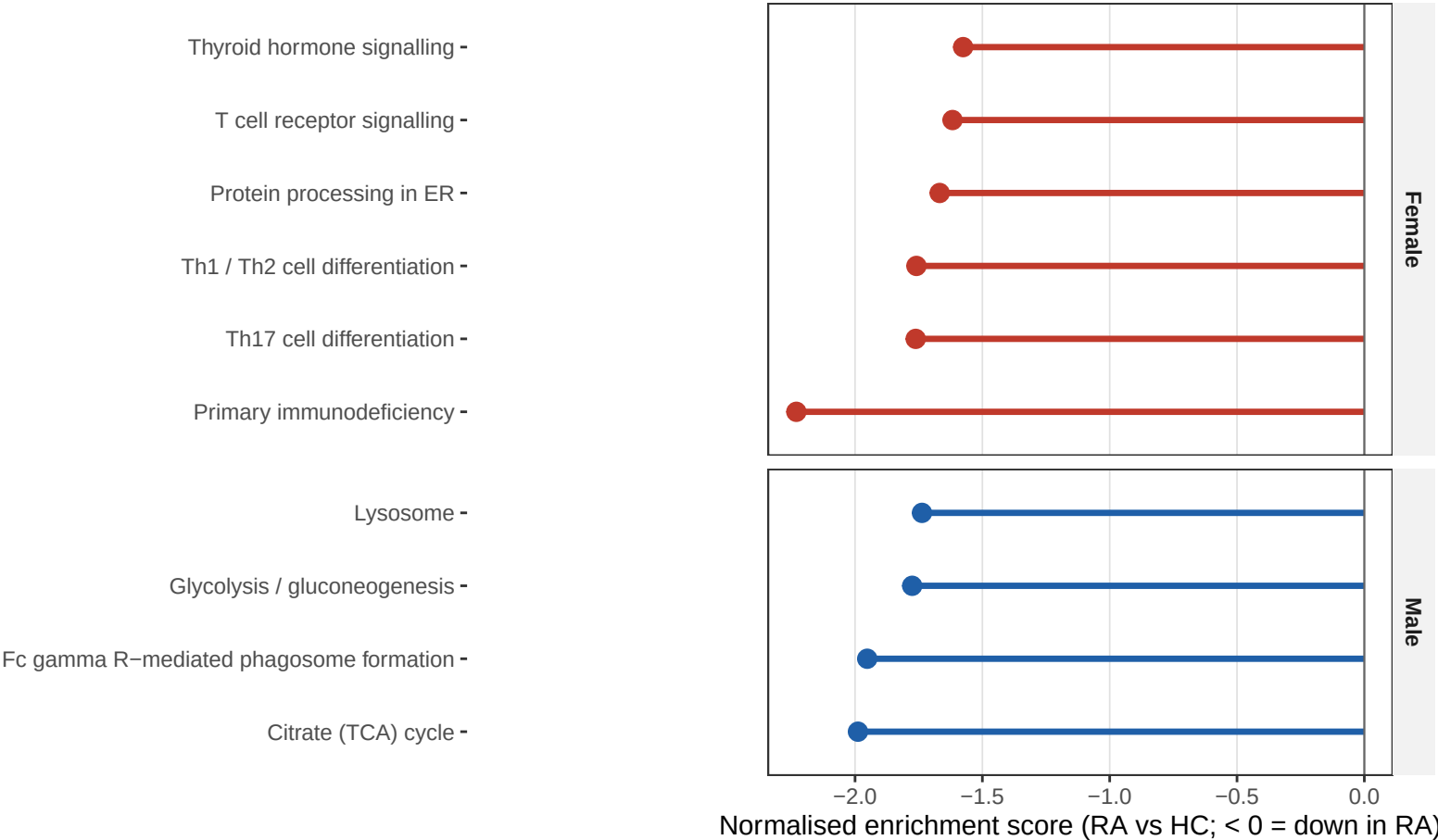
A Shared core: enriched in both sexes
KEGG pathways significant in female and male RA DEG sets



B Sex-restricted: enriched in one sex only
Female-only left, male-only right; NES where GSEA also significant



C Ranked GSEA: displaced in one sex only
Curated pathways with FDR < 0.05 in exactly one sex



D MR-prioritised causal genes
KEGG enrichment of each sex's candidate set (expressed-gene background)

